

Q. ATM Simulation

The image shows a screenshot of an online Python compiler interface, specifically the Programiz Python Online Compiler. The browser address bar shows the URL <https://www.programiz.com/python-programming/online-compiler/>. The interface is divided into two main sections: a code editor on the left and an output window on the right.

Code Editor (main.py):

```
1 class ATM:
2     def __init__(self, balance=1000):
3         self.balance=balance
4     def check_balance(self):
5         print(f"Your balance is: ${self.balance}")
6     def deposit(self, amount):
7         self.balance+=amount
8         print(f"Deposited: ${amount}")
9     def withdraw(self, amount):
10        if amount > self.balance:
11            print("Insufficient Balance!")
12        else:
13            self.balance-=amount
14            print(f"Withdrawn: ${amount}")
15
16 def main():
17     atm=ATM()
18     while True:
19         print("\n1. Check Balance\n2. Deposit\n3. Withdraw\n4. Exit")
20         choice=input("Enter choice:")
21
22         if choice=="1":
23             atm.check_balance()
24         elif choice=="2":
25             amt=float(input("Enter deposit amount:"))
26             atm.deposit(amt)
27         elif choice=="3":
28             amt=float(input("Enter withdrawal amount:"))
29             atm.withdraw(amt)
30         elif choice=="4":
31             print("Thank you using our services!")
32             break
33         else:
34             print("Invalid choice! Try again.")
35
36
```

Output Window:

The output window shows the results of the program execution. It displays a menu with options 1 (Check Balance), 2 (Deposit), 3 (Withdraw), and 4 (Exit). The user enters choice 1, and the program outputs "Your balance is: \$1000". The user then enters choice 2, and the program prompts for a deposit amount. The user enters 1234, and the program outputs "Deposited: \$1234.0". The user then enters choice 3, and the program prompts for a withdrawal amount. The user enters 2000, and the program outputs "Withdrawn: \$2000.0". The user then enters choice 1, and the program outputs "Your balance is: \$234.0".

The browser's taskbar at the bottom shows the system clock as 15:21 on 20-02-2025, and the weather as 89°F Haze.

Q. Student Grading System

The screenshot displays the Programiz Python Online Compiler interface. The left pane shows the code for a Student Grading System, and the right pane shows the output of the program.

Code (main.py):

```
1 class GradeSystem:
2     def __init__(self):
3         self.grades={}
4     def add_grade(self,name, grade):
5         self.grades[name]=grade
6         print(f"Added:{name}={grade}")
7     def view_grades(self):
8         if not self.grades:
9             print("No grades available")
10        else:
11            print("\nStudent Grades:")
12            for name, grade in self.grades.items():
13                print(f"{name}:{grade}")
14    def calculate_average(self):
15        if not self.grades:
16            print("No grades available")
17        else:
18            avg=sum(self.grades.values())/len(self.grades)
19            print(f"Class Average: {avg:.2f}")
20
21 def main():
22     system = GradeSystem()
23     while True:
24         print("\n1.Add Grade\n2.View Grade\n3.Calculate
25         Average\n4.Exit")
26         choice=input("Enter choice:")
27         if choice == "1":
28             name = input("Enter student name:")
29             grade=float(input("Enter grade:"))
30             system.add_grade(name,grade)
31         elif choice=="2":
32             system.view_grades()
33         elif choice=="3":
34             system.calculate_average()
35         elif choice=="4":
36             print("Exiting Grade System")
37             break
38         else:
39             print("Invalid choice")
40     main()
```

Output:

```
1.Add Grade
2.View Grade
3.Calculate Average
4.Exit
Enter choice:1
Enter student name:KP
Enter grade:90
Added: KP=90.0

1.Add Grade
2.View Grade
3.Calculate Average
4.Exit
Enter choice:1
Enter student name:KP
Enter grade:80
Added: KP=80.0

1.Add Grade
2.View Grade
3.Calculate Average
4.Exit
Enter choice:3
Class Average: 80.00

1.Add Grade
2.View Grade
3.Calculate Average
4.Exit
Enter choice:4
Exiting Grade System

=== Code Execution Successful ===
```

The browser window shows the URL <https://www.programiz.com/python-programming/online-compiler/> and the date 20-02-2025.

Q. Hospital Management System

The screenshot displays the Programiz Python Online Compiler interface. The left pane shows the code for a Hospital Management System, and the right pane shows the output of the program.

Code (main.py):

```
1 class Hospital:
2     def __init__(self):
3         self.patients={}
4     def add_patient(self,id,name,age,disease):
5         self.patients[id]={"Name":name,"Age":age,"Disease":disease}
6         print(f"Patient {name} added!")
7     def view_patients(self):
8         if not self.patients:
9             print("No patients registered!")
10        else:
11            print("\nPatient Records:")
12            for id,details in self.patients.items():
13                print(f"ID: {id} - {details}")
14    def remove_patient(self,id):
15        if id in self.patients:
16            del self.patients[id]
17            print("Patient removed")
18        else:
19            print("Patient not found!")
20    def main():
21        hospital =Hospital()
22        while True:
23            print("\n1.Add Patient\n2.View Patients\n3.Remove Patient\n4.Exit")
24            choice=input("Enter choice:")
25            if choice == "1":
26                id=input("Enter Patient ID:")
27                name=input("Enter Name:")
28                age=input("Enter Age:")
29                disease=input("Enter Disease:")
30                hospital.add_patient(id,name,age,disease)
31            elif choice=="2":
32                hospital.view_patients()
33            elif choice == "3":
34                id=input("Enter Patient ID to remove:")
35                hospital.remove_patient(id)
36            elif choice == "4":
37                print("Exiting Hospital System")
38                break
39            else:
40                pass
```

Output:

```
1.Add Patient
2.View Patients
3.Remove Patient
4.Exit
Enter choice:1
Enter Patient ID:12AB
Enter Name:KP
Enter Age:22
Enter Disease:Sore Throat
Patient KP added!

1.Add Patient
2.View Patients
3.Remove Patient
4.Exit
Enter choice:2
Patient Records:
ID: 12AB - {'Name': 'KP', 'Age': '22', 'Disease': 'Sore Throat'}
```

The browser window shows the URL <https://www.programiz.com/python-programming/online-compiler/>. The system tray at the bottom indicates a temperature of 89°F, weather of Haze, and the date 20-02-2025.

Q. Order Management System

The screenshot shows a web browser window with the URL <https://www.programiz.com/python-programming/online-compiler/>. The browser tabs include "Techademy", "online python compiler - Search", "Online Python Compiler (Interpr...", and "main.py".

The Programiz Python Online Compiler interface is displayed. It has a dark theme and includes a "Programiz PRO" button in the top right corner. The main area is divided into two panels: a code editor on the left and an output console on the right.

The code editor shows the following Python code in `main.py`:

```
1 class Product:
2     def __init__(self, name, price):
3         self.name = name
4         self.price = price
5
6 class ShoppingCart:
7     def __init__(self):
8         self.cart = []
9     def add_product(self, product):
10        self.cart.append(product)
11        print(f"{product.name} added to cart")
12    def view_cart(self):
13        if not self.cart:
14            print("Cart is empty")
15        else:
16            print("\n Shopping Cart")
17            total = 0
18            for p in self.cart:
19                print(f"- {p.name}: ${p.price}")
20                total += p.price
21            print(f"Total: ${total}")
```

The output console shows the results of running the code. It displays a menu of items to add to the cart, followed by the user's choice to add a laptop, and then the updated cart contents and total price.

Output:

```
1. Add Laptop ($1000)
2. Add Headphones($150)
3. Add Mouse($50)
4. View Cart
5. Checkout
6. Exit
Enter choice: 4
Cart is empty

1. Add Laptop ($1000)
2. Add Headphones($150)
3. Add Mouse($50)
4. View Cart
5. Checkout
6. Exit
Enter choice: 1
Laptop added to cart

1. Add Laptop ($1000)
2. Add Headphones($150)
```

The bottom of the image shows a Windows taskbar with the date and time set to 15:24 on 20-02-2025, and the language set to ENG IN.

Programiz Python Online Compiler

main.py

22- def checkout(self):
23- if not self.cart:
24- print("Cart is empty")
25- else:
26- self.view_cart()
27- print("Proceeding to checkout...")
28-
29- def main():
30- cart=ShoppingCart()
31- products = {
32- "1":Product("Laptop",1000),
33- "2":Product("Headphones",150),
34- "3":Product("Mouse",50),
35- }
36-
37- while True:
38- print("\n1. Add Laptop (\$1000) \n2. Add Headphones(\$150)
 \n3. Add Mouse(\$50)\n4. View Cart\n5.Checkout\n6.Exit")
39- choice=input("Enter choice:")
40-

Output

Enter choice:2
Headphones added to cart

1. Add Laptop (\$1000)
2. Add Headphones(\$150)
3. Add Mouse(\$50)
4. View Cart
5.Checkout
6.Exit
Enter choice:4

Shopping Cart
- Laptop: \$1000
- Headphones: \$150
Total: \$1150

1. Add Laptop (\$1000)
2. Add Headphones(\$150)
3. Add Mouse(\$50)
4. View Cart

Programiz Python Online Compiler

main.py

36-
37- while True:
38- print("\n1. Add Laptop (\$1000) \n2. Add Headphones(\$150)
 \n3. Add Mouse(\$50)\n4. View Cart\n5.Checkout\n6.Exit")
39- choice=input("Enter choice:")
40-
41- if choice in products:
42- cart.add_product(products[choice])
43- elif choice == "4":
44- cart.view_cart()
45- elif choice == "5":
46- cart.checkout()
47- elif choice == "6":
48- print("Thank you for shopping!")
49- break
50- else:
51- print("Invalid Choice!")
52-
53- main()

Output

5. Checkout
6.Exit
Enter choice:5

Shopping Cart
- Laptop: \$1000
- Headphones: \$150
Total: \$1150
Proceeding to checkout...

1. Add Laptop (\$1000)
2. Add Headphones(\$150)
3. Add Mouse(\$50)
4. View Cart
5. Checkout
6.Exit
Enter choice:6
Thank you for shopping!

=== Code Execution Successful ===